Thermodynamics — 1-Hour Refresher (with Explanations)

1) First Law of Thermodynamics

The First Law states that energy is conserved. The change in internal energy equals the heat supplied to the system minus the work done by the system.

Equation (finite form): $\Delta U = Q - W$

This means if a system absorbs heat Q and does work W on its surroundings, its internal energy U changes by ΔU.

Differential form (for reversible processes): $dU = T\,dS - P\,dV$

This differential form links internal energy to entropy (S) and volume (V).

2) Second Law of Thermodynamics

The Second Law introduces the concept of entropy and the direction of spontaneous processes. Heat flows naturally from hot to cold.

Entropy change (reversible): $dS = \delta Q_{rev} / T$

Entropy S increases when heat δQ is absorbed at temperature T. In isolated systems, total entropy never decreases.

3) Heat Capacities

Heat capacity measures how much heat is required to raise temperature. At constant volume and pressure:

$C_V = (\partial U/\partial T)_V$, $C_P = (\partial H/\partial T)_P$

For ideal gases, U depends only on T, so $\Delta U = n\,C_V\,\Delta T$, and enthalpy H depends only on T, so $\Delta H = n\,C_P\,\Delta T$.